Unit #5
Chapter #10.3:
Tools for High Frequency Response Analysis
Lecture 28

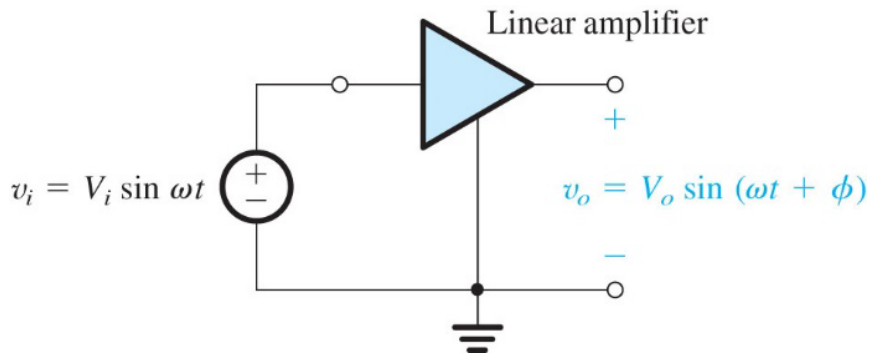
from **Microelectronic Circuits** Text

by Sedra and Smith

Oxford Publishing

10.3 **High-frequency Gain Function**

- **The linear amplifier gain can always be expressed as a function of the complex frequency, s.**



$$A(s) = \frac{x_o(s)}{x_{sig}(s)} = A_M F_H(s)$$

- **Here A_M is the maximum (midband) gain and F_H is the frequency dependent variable**

$$F_H(s) = \frac{(1 + s/\omega_{z1})(1 + s/\omega_{z1}) \cdots (1 + s/\omega_{zn})}{(1 + s/\omega_{p1})(1 + s/\omega_{p1}) \cdots (1 + s/\omega_{pm})}$$

ω_{zi} **are zeros**

ω_{pj} **are poles**

10.3 **Estimating the 3-dB frequency f_H**

- **Given the potential for such a complicated frequency function, we would like some simple tools that can help us quickly estimate f_H … given reasonable assumptions**

$$F_H(s) \approx \frac{1}{1 + s/\omega_H}$$

This would be the dominant pole

- **By definition if $\omega = \omega_H$**

$$\left|F_H(\omega_H)\right|^2 \approx \frac{1}{2}$$

10.3 **Estimating the 3-dB frequency f_H**

- **Let us consider the case when there are two zeros and two poles**

$$F_H(s) = \frac{(1+s/\omega_{z1})(1+s/\omega_{z2})}{(1+s/\omega_{p1})(1+s/\omega_{p2})} \approx \frac{1}{(1+s/\omega_H)}$$

- **Then**

$$|F_H(\omega_H)|^2 = \frac{(1+\omega_H{}^2/\omega_{z1}{}^2)(1+\omega_H{}^2/\omega_{z2}{}^2)}{(1+\omega_H{}^2/\omega_{p1}{}^2)(1+\omega_H{}^2/\omega_{p2}{}^2)} \approx \frac{1}{2}$$

so

$$\frac{1+\omega_H{}^2\left(\frac{1}{\omega_{z1}{}^2}+\frac{1}{\omega_{z2}{}^2}\right)+\omega_H{}^4/\omega_{z1}{}^2\omega_{z2}{}^2}{1+\omega_H{}^2\left(\frac{1}{\omega_{p1}{}^2}+\frac{1}{\omega_{p2}{}^2}\right)+\omega_H{}^4/\omega_{p1}{}^2\omega_{p2}{}^2} \approx \frac{1}{2}$$

10.3 **Estimating the 3-dB frequency f_H**

- **Now ω_H is usually smaller than all the high frequency poles and zeros so we can neglect the ω_H^4 term**

$$\frac{1+\omega_H^{\,2}\left(\dfrac{1}{\omega_{z1}^{\,2}}+\dfrac{1}{\omega_{z2}^{\,2}}\right)+\omega_H^{\,4}\,\omega_{z1}^{\,2}\,\omega_{z2}^{\,2}}{1+\omega_H^{\,2}\left(\dfrac{1}{\omega_{p1}^{\,2}}+\dfrac{1}{\omega_{p2}^{\,2}}\right)+\omega_H^{\,4}\,\omega_{p1}^{\,2}\,\omega_{p2}^{\,2}}\approx\frac{1}{2}$$

- **Then**

$$1+\omega_H^{\,2}\left(\frac{1}{\omega_{z1}^{\,2}}+\frac{1}{\omega_{z2}^{\,2}}\right)\approx\frac{1}{2}\left[1+\omega_H^{\,2}\left(\frac{1}{\omega_{p1}^{\,2}}+\frac{1}{\omega_{p2}^{\,2}}\right)\right]$$

Solve the above for $\boldsymbol{\omega}_{\mathrm{H}}$

$$\omega_H\approx 1\Big/\sqrt{\frac{1}{\omega_{p1}^{\,2}}+\frac{1}{\omega_{p2}^{\,2}}-\frac{2}{\omega_{z1}^{\,2}}-\frac{2}{\omega_{z2}^{\,2}}}$$

10.3 Estimating the 3-dB frequency f_H (Square Root Method)

- **The previous relationship can be extended to any number of poles and zeros as**

$$\omega_H \approx 1 \Big/ \sqrt{\left(\frac{1}{\omega_{p1}^{\ 2}} + \frac{1}{\omega_{p2}^{\ 2}} + \cdots\right) - 2\left(\frac{1}{\omega_{z1}^{\ 2}} + \frac{1}{\omega_{z2}^{\ 2}} + \cdots\right)}$$

- **Note that if one of the poles is dominant, e.g., ω_{p1} << all other poles and zeros**

$$\omega_H \approx \omega_{p1}$$

10.3 **Example**

The high frequency response of an amplifier is characterized by the transfer function

$$F_H(s) = \frac{1 - s/10^5}{\left(1 + s/10^4\right)\left(1 + s/4x10^4\right)}$$

Use the above square root method to estimate the 3-dB frequency